Improving Physics-Informed Neural Networks for Seismic Wave Propagation and Full Waveform Inversion

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1 Introduction

Physics-informed neural networks (PINNs) (Raissi et al., 2019) embed governing PDEs directly into a neural network's loss function, enabling both forward and inverse problems to be solved in a unified, meshless optimization framework. Rasht-Behesht et al. (2022) applied PINNs to the 2D acoustic wave equation for full waveform inversion (FWI), recovering subsurface wavespeed distributions from sparse surface seismograms and two early-time wavefield snapshots. Their results are promising, but the methodology relies on hand-tuned loss weights, makes unjustified choices between activation functions across different test cases, and acknowledges but does not address the well-known spectral bias of neural networks near sharp material interfaces. We reproduce the ellipsoidal anomaly inverse problem (Case 3) from Rasht-Behesht et al. (2022) and systematically evaluate four improvements:

1. Adaptive loss weighting based on gradient statistics (Wang et al., 2021),
2. Activation function comparison: tanh, sinusoidal/SIREN (Sitzmann et al., 2020), and trainable adaptive activations (Jagtap et al., 2020),
3. Random Fourier feature input encoding for spectral bias mitigation (Tancik et al., 2020),
4. Residual-based adaptive collocation point resampling.

We replace the original SpecFem2D ground-truth solver with a self-contained finite-difference code for full reproducibility. The scalar acoustic wave equation in a 2D medium with negligible density variations is

$$\alpha^2(x,z)\,\nabla^2\phi = \frac{\partial^2\phi}{\partial t^2}, \tag{1}$$

where $\phi(x,z,t)$ is the scalar wave potential, $\alpha(x,z)$ is the spatially varying P-wave velocity, and $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial z^2$. For the forward problem, α is known and one solves for ϕ; for the inverse problem, α must be inferred from observed waveforms at sparse receiver locations. A PINN approximates ϕ with a fully connected network $\hat{\phi}_\Theta(x,z,t)$. The PDE is enforced at randomly sampled collocation points by penalizing the residual via automatic differentiation. The total loss combines four terms:

$$\mathcal{L}(\Theta) = \lambda_{\text{pde}}\mathcal{L}_{\text{pde}} + \lambda_{\text{snap}}\mathcal{L}_{\text{snap}} + \lambda_{\text{obs}}\mathcal{L}_{\text{obs}} + \lambda_{\text{fs}}\mathcal{L}_{\text{fs}}, \tag{2}$$

where $\mathcal{L}_{\text{pde}}$ is the mean squared PDE residual, $\mathcal{L}_{\text{snap}}$ fits two early-time wavefield snapshots encoding the source, $\mathcal{L}_{\text{obs}}$ matches surface seismograms, and $\mathcal{L}_{\text{fs}}$ enforces the stress-free condition at $z = 0$. A second network $\hat{\alpha}_\Psi(x,z)$ parameterizes the unknown wavespeed; both networks are trained jointly.

2 Methodology

2.1 Ground Truth: Finite Difference Solver

We generate all training data using a custom 2D finite-difference solver in PyTorch with second-order central differences and explicit leapfrog time stepping ($\Delta t \leq \Delta x/(\alpha_{\max}\sqrt{2})$ for CFL stability). Absorbing sponge layers on the left, right, and bottom boundaries prevent artificial reflections; a stress-free condition at $z = 0$ is applied via the image method. The source is a 20 Hz Ricker wavelet matching Case 3 of Rasht-Behesht et al. (2022). Two snapshots are taken at $t_1 = 0.100$ s and $t_2 = 0.115$ s, and 20 equally spaced surface receivers record 0.4 s of data.

2.2 PINN Architecture

The inverse problem employs two networks trained jointly:

- **WavefieldNet** $(x, z, t) \to \phi$: 4 hidden layers of 50 neurons, tanh activation (baseline).
- **WavespeedNet** $(x, z) \to \alpha$: 4 hidden layers of 20 neurons, tanh activation, with output $\hat{\alpha} = \alpha_{\text{bg}} + \Delta\alpha \cdot \tanh(\text{raw}) \cdot m(x, z)$, where $\alpha_{\text{bg}} = 3.0$ km/s, $\Delta\alpha = 1.0$ km/s, and $m(x, z)$ is a smooth mask restricting inversion to the domain interior.

All inputs are normalized to $[-1, 1]$. The PDE residual accounts for the chain-rule scaling:

$$\alpha^2 \left(s_x^2 \frac{\partial^2 \phi}{\partial \hat{x}^2} + s_z^2 \frac{\partial^2 \phi}{\partial \hat{z}^2} \right) = s_t^2 \frac{\partial^2 \phi}{\partial \hat{t}^2}, \tag{3}$$

where $s_x = 2/L_x$, $s_z = 2/L_z$, $s_t = 2/L_t$. Omitting these factors causes convergence to an incorrect solution, a subtlety not discussed in Rasht-Behesht et al. (2022).

2.3 Proposed Improvements

2.3.1 Improvement 1: Adaptive Loss Weighting

We implement the gradient-balancing algorithm of Wang et al. (2021), which adjusts weights dynamically:

$$\hat{\lambda}_i = \frac{\max_j \|\nabla_\Theta \mathcal{L}_j\|_\infty}{\|\nabla_\Theta \mathcal{L}_i\|_\infty}, \qquad \lambda_i^{(k)} = (1 - \beta)\, \lambda_i^{(k-1)} + \beta\, \hat{\lambda}_i, \tag{4}$$

with $\beta = 0.1$.

2.3.2 Improvement 2: Activation Function Study

Rasht-Behesht et al. (2022) use sin activations for some cases and tanh for others without justification. We compare: (a) **tanh** (baseline), (b) **sin** (SIREN, $\omega_0 = 30$) following Sitzmann et al. (2020), and (c) **adaptive tanh** of the form $\tanh(n \cdot a \cdot x)$ with trainable a initialized to 1.0 and $n = 5$, following Jagtap et al. (2020).

2.3.3 Improvement 3: Fourier Feature Encoding

Following Tancik et al. (2020), we apply a random Fourier feature mapping:

$$\gamma(\mathbf{v}) = \big[\cos(2\pi \mathbf{B}\mathbf{v}),\ \sin(2\pi \mathbf{B}\mathbf{v})\big], \qquad \mathbf{B} \sim \mathcal{N}(0, \sigma^2), \tag{5}$$

with $\sigma = 4.0$ and 64 features (mapping 3D input to 128 dimensions).

2.3.4 Improvement 4: Residual-Based Adaptive Collocation

Every 500 epochs, we evaluate the PDE residual on 200,000 candidate points and resample $N_{\text{pde}} = 10{,}000$ collocation points: 50% with probability proportional to the local residual magnitude, 50% uniform.

3 Experimental Setup

All experiments use the same inverse problem: an ellipsoidal anomaly ($\alpha = 2.5$ km/s) in a 3.0 km/s background on a 1.4×0.5 km domain, matching Case 3 of Rasht-Behesht et al. (2022). Each experiment modifies exactly one component relative to the baseline. All runs use identical random seeds and are trained for 30,000 epochs. We report wavespeed RMSE and relative L^2 displacement error at $t = 0.250$ s. All training was performed on a single NVIDIA RTX 4070 GPU. Hyperparameters are listed in Table 1.

Table 1: Training hyperparameters (shared across all experiments unless noted).

Parameter	Value
Learning rate	10^{-3} (cosine decay $\to 10^{-5}$)
Optimizer	Adam
Epochs	30,000
N_{pde} / N_{snap} / N_{obs} / N_{fs}	10,000 / 3,600 / 13,580 / 5,000
Gradient clipping	$\|\cdot\|_2 \leq 1.0$
Baseline $(\lambda_{\text{pde}}, \lambda_{\text{snap}}, \lambda_{\text{fs}}, \lambda_{\text{obs}})$	$(0.1, 1.0, 0.1, 1.0)$

The baseline loss weights were adopted from Rasht-Behesht et al. (2022).

4 Results

4.1 Baseline Reproduction

The baseline PINN successfully identifies the anomaly location, achieving a wavespeed RMSE of 0.190 km/s (Figure 1). The velocity reduction is underestimated (∼2.82 km/s predicted vs. 2.5 km/s true) and the boundary is smoothed. The wavefield captures the general propagation structure but with large amplitude errors (relative error 0.875), particularly behind the wavefront where scattering produces complex interference.

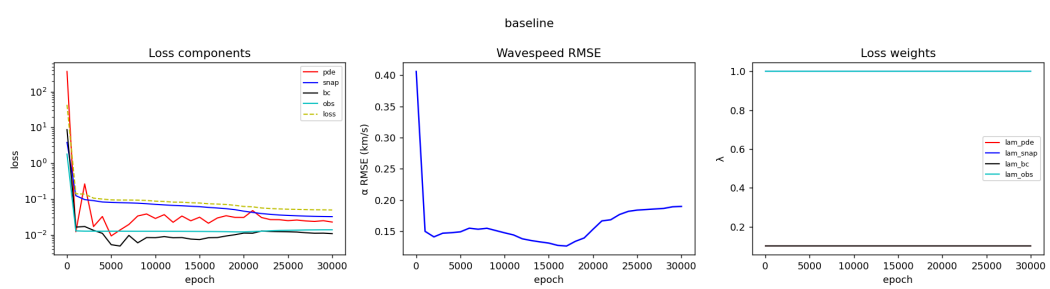

Figure 1: Baseline results. Top: true wavespeed (left), predicted (center), absolute error (right; RMSE = 0.190 km/s). Bottom: FD wavefield at $t = 0.250$ s (left), PINN prediction (center), error (right; rel. err = 0.875).

4.2 Effect of Individual Improvements

Table 2 summarizes all experiments. The two most effective improvements are sinusoidal activations and adaptive collocation, each reducing wavespeed RMSE by ∼34%.

Table 2: Quantitative results (30,000 epochs). Bold: best in column (excluding failures).

Experiment	α RMSE (km/s)	$\lvert\Delta\mathbf{U}\rvert$ rel. err	Time (s)
Baseline (tanh, fixed λ)	0.190	0.875	1212
Adaptive weights	0.187	0.898	1224
Sin activation (SIREN)	**0.126**	0.909	1209
Adaptive tanh activation	1.119	1.064	1298
Fourier features (σ=4)	0.229	0.990	1340
Adaptive collocation	**0.125**	1.026	1162

Adaptive loss weighting provides marginal benefit (RMSE 0.187 vs. 0.190). The dynamically computed λ_{obs} grows to ∼7,000 by end of training, indicating a large gradient imbalance between the observation and PDE losses, but the hand-tuned baseline weights already approximately balance gradient magnitudes for this configuration. **Sinusoidal activations** reduce the RMSE to 0.126 km/s (Figure 2), with the minimum predicted velocity reaching ∼2.74 km/s, closer to the true 2.5 km/s than the baseline's 2.82 km/s.

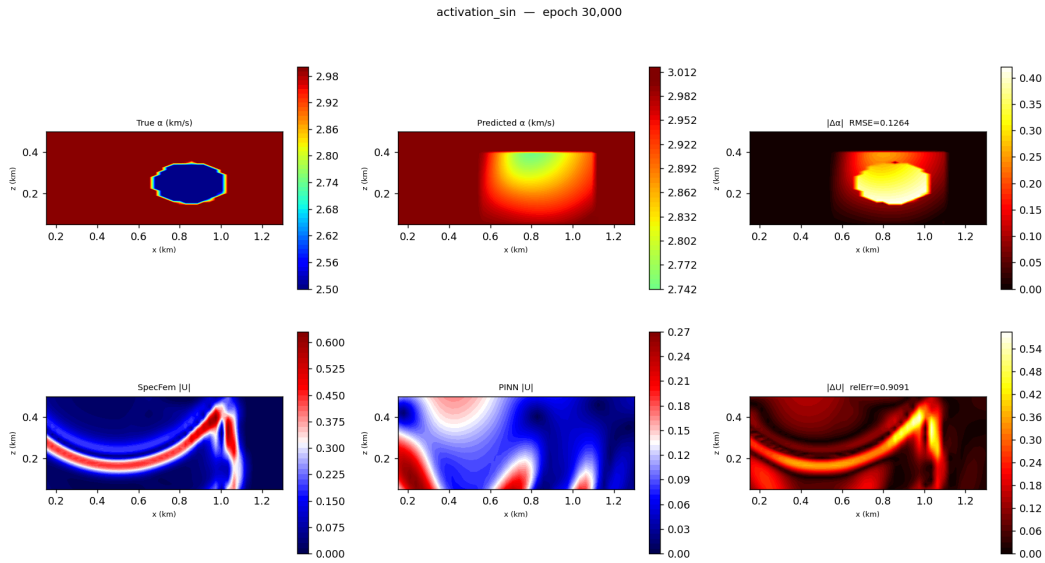

Figure 2: Sinusoidal activation results (RMSE = 0.126 km/s). The anomaly is better resolved with a more clearly defined low-velocity region.

Adaptive tanh activations fail catastrophically (RMSE = 1.119 km/s). The wavespeed network collapses to a nearly uniform ∼1.0 km/s prediction across the domain (Figure 3). The failure mechanism can be understood quantitatively. Since inputs are normalized to $[-1, 1]$, a typical pre-activation argument has magnitude $|n \cdot a \cdot x| \approx n|a|$. For $n = 5$, once a exceeds approximately 0.6 the peak argument reaches 3.0, where $\tanh'(3.0) \approx 0.01$, a 100× gradient attenuation per layer. With 4 hidden layers, the effective gradient signal reaching the early layers is attenuated by a factor of roughly $0.01^4 = 10^{-8}$, making meaningful weight updates impossible. Because a is updated by the same optimizer as the network weights,

it grows during the early high-loss phase when gradients are large and noisy; once the tanh arguments saturate, the vanishing gradients prevent a from recovering. The coupled inverse problem exacerbates this: the wavespeed network's collapse feeds an incorrect velocity into the PDE residual, creating a feedback loop that locks both networks into their failed states. This pathology does not arise in the forward-only problems studied in Jagtap et al. (2020), where the target solution is fixed and cannot co-evolve with the activation parameters.

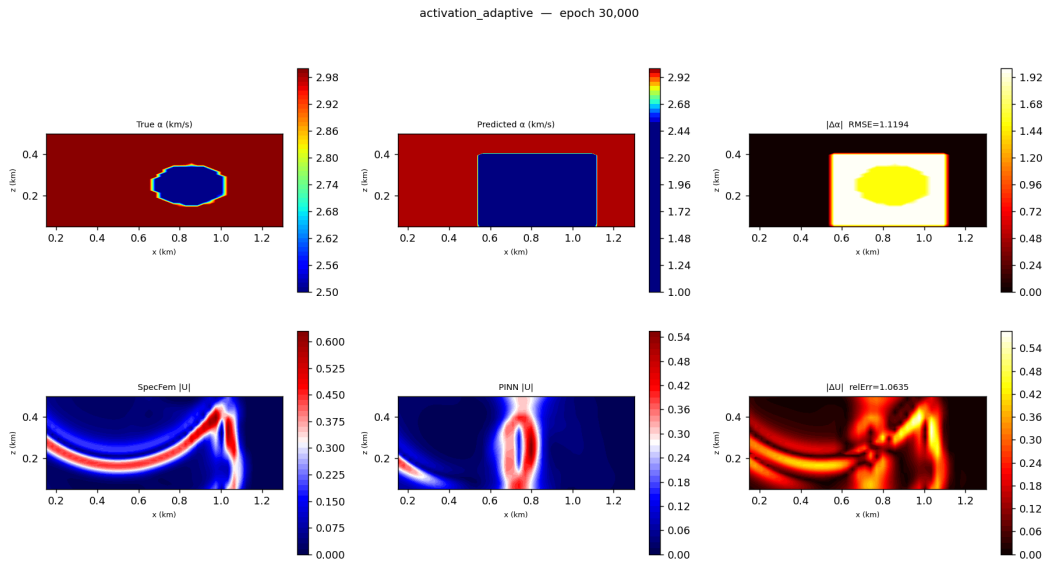

Figure 3: Adaptive tanh failure. The scaling parameters saturate early, locking the wavespeed network into a uniform low-velocity prediction (RMSE = 1.119 km/s).

Fourier feature encoding degrades velocity recovery (RMSE = 0.229 km/s). The wavefield shows pronounced striped artifacts at amplitudes $\sim 10^{-3}$, orders of magnitude below the ground truth (Figure 4). The RFF encoding with $\sigma = 4.0$ places peak sensitivity at $\sim$2.9 cycles/km, a factor of two below the wavefield's dominant spatial frequency of $\sim$6.7 cycles/km ($\alpha/f = 3.0/20 = 0.15$ km). This mismatch amplifies representational capacity in an irrelevant frequency band while allowing the network to fit spurious oscillations consistent with sparse observations but violating the PDE. Section 5.2 develops this argument further.

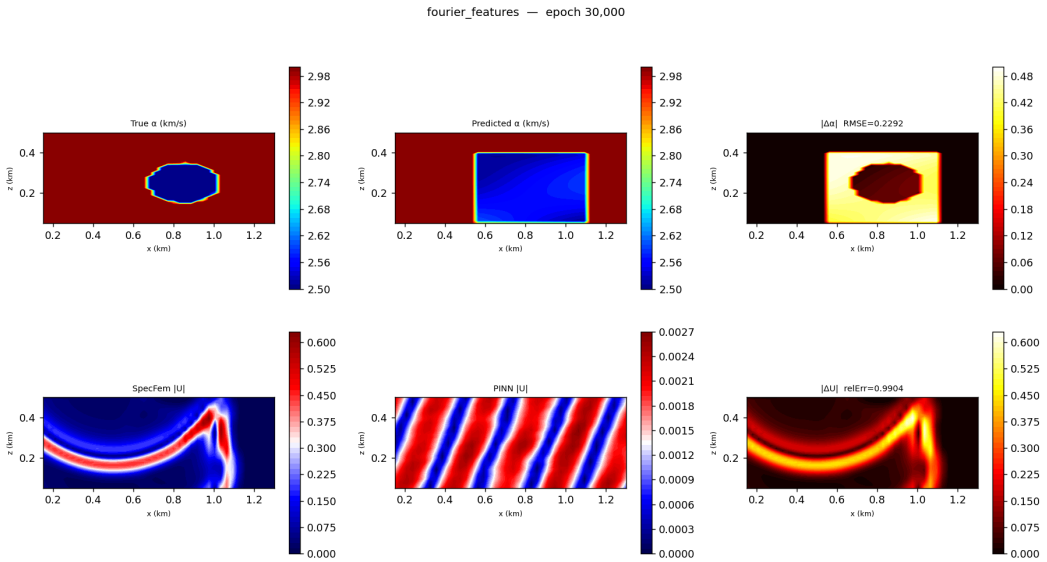

Figure 4: Fourier feature results. The anomaly shape is sharper but the wavefield is dominated by spurious oscillations at $\sim 10^{-3}$ amplitude.

Adaptive collocation achieves the best RMSE (0.125 km/s) with the shortest training time (1162 s, Figure 5), as concentrating points near high-residual regions enforces the PDE more efficiently. However, the wavefield relative error (1.026) is higher than the baseline, a tradeoff explored in Section 5.3.

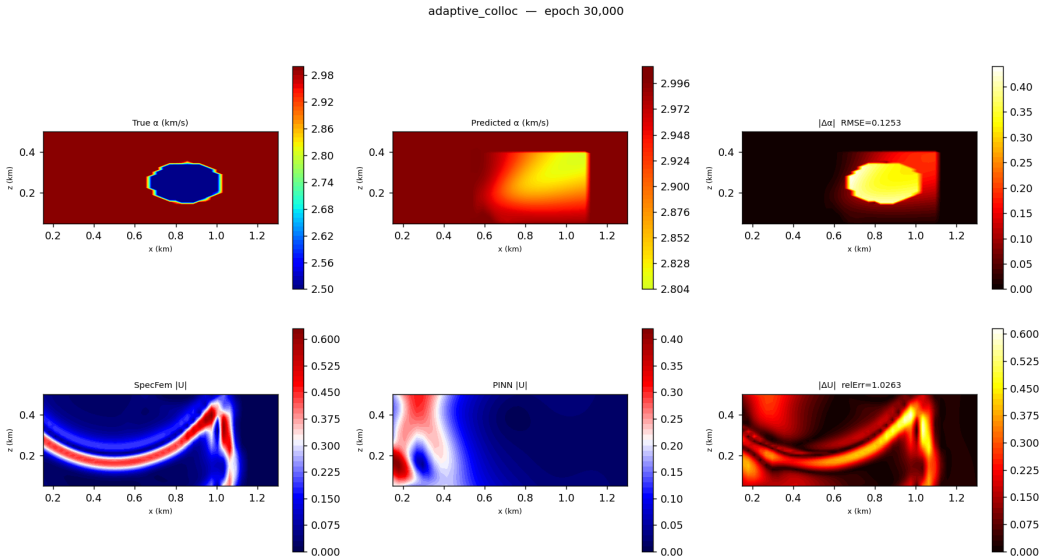

Figure 5: Adaptive collocation results (RMSE = 0.125 km/s).

4.3 Convergence

Figure 6 compares training loss and RMSE trajectories. Successful runs converge rapidly by $\sim$10,000 epochs; the adaptive activation and Fourier feature runs plateau at substantially higher loss values.

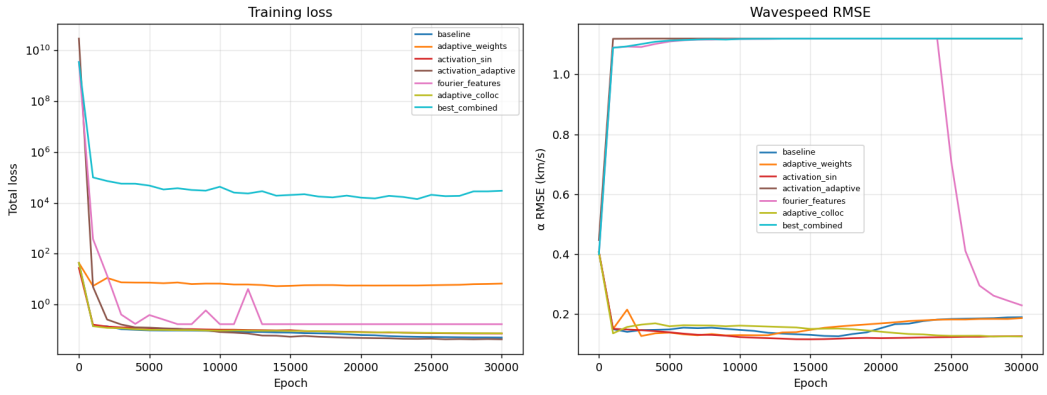

Figure 6: Training loss (left, log scale) and wavespeed RMSE (right) vs. epoch.

5 Discussion

5.1 Comparison with the Original Study

Direct quantitative comparison with Rasht-Behesht et al. (2022) is difficult: the original reports Case 3 results qualitatively rather than as numerical RMSE values, and several setup differences (SpecFem2D with PML vs. our FD solver with sponge layers; 5-layer sin-activation network trained for 400,000 iterations with Adam + L-BFGS vs. our 4-layer tanh baseline at 30,000 Adam epochs) preclude matching absolute numbers. The original notes that a smaller network "still yields a good estimate of the wavespeed distribution, however the recovered wavefield is less accurate," which is qualitatively consistent with our baseline (RMSE = 0.190 km/s, wavefield relative error 0.875). Our absolute values should therefore not be read as a reproduction of the original's performance. The baseline instead establishes a controlled reference against which improvements are evaluated under identical conditions; the relative rankings, which are the primary contribution, are internally valid provided the improvements interact with problem structure rather than solver-specific artifacts.

5.2 Spectral Interpretation of Activation and Encoding Effects

The contrasting outcomes can be understood through the neural tangent kernel (NTK) (Jacot et al., 2018). The NTK eigenvalue spectrum controls which frequency components are learned first; for tanh networks, eigenvalues decay rapidly with frequency, producing spectral bias toward low-frequency solutions. Sinusoidal activations introduce periodic structure into $\nabla_\Theta f$, flattening the eigenvalue decay and allowing high-frequency components to be learned at comparable rates. This explains the improved wavespeed recovery at the anomaly boundary, where spatial frequency content is $\sim$5.6 cycles/km. Fourier feature encoding achieves a similar modification but with less control: the RFF bandwidth set by $\sigma = 4.0$ places peak sensitivity at $\sim$2.9 cycles/km, misaligned with both the anomaly scale and the wavefield's dominant spatial frequency ($\sim$6.7 cycles/km). A principled choice $\sigma \approx fL_x/\alpha = 9.3$ would better match the wavefield spectrum. Adaptive collocation does not modify spectral properties but concentrates the PDE constraint near high-residual regions (wavefronts, material interfaces), achieving a similar practical effect through a complementary mechanism operating on the loss landscape rather than the architecture.

5.3 The Velocity–Wavefield Error Gap

A striking pattern in Table 2 is that every run has wavefield relative error above 0.87, and the two best velocity runs actually produce *worse* wavefield errors than the baseline. This reflects a structural tension in the coupled optimization: the wavefield network must simultaneously match observations and

satisfy the PDE with whatever velocity the wavespeed network currently predicts. Early in training the velocity is poor, so the network learns a compromise; as velocity improves, the wavefield must unlearn this compromise by traversing a high-loss region. Improvements that accelerate velocity convergence (sin activations, adaptive collocation) exacerbate this: the velocity changes faster than the wavefield can track. This suggests the error is a training dynamics issue rather than a capacity limitation, and that staged training, involving first fitting the forward problem and then gradually introducing the velocity as learnable, might help.

5.4 Limitations

Our second-order FD solver may introduce numerical dispersion absent from the original's spectral element solver. The networks are small (4 layers, 50 neurons) relative to Rasht-Behesht et al. (2022); improvement effectiveness may change at scale.

We test a single synthetic configuration and train for 30,000 epochs versus up to 400,000 in the reference. With each run requiring 20–23 minutes on a single NVIDIA RTX 4070 (Table 2), a combinatorial sweep over improvements and hyperparameters was not feasible within our compute budget.

6 Conclusion

Among four proposed improvements to the PINN-based FWI framework of Rasht-Behesht et al. (2022), sinusoidal activations and adaptive collocation each reduce wavespeed RMSE by $\sim$34%, while adaptive loss weighting provides minimal benefit and Fourier features degrade performance. Adaptive tanh activations fail due to a saturation feedback loop specific to coupled inverse problems. The most unexpected finding is that better velocity recovery does not imply better wavefield reconstruction. The velocity–wavefield error gap suggests that the coupled optimization has fundamentally different dynamics than a pure forward PINN: the wavefield network must track a moving target as the velocity evolves, and faster velocity convergence can actually destabilize this tracking. For practitioners starting a PINN-based inversion project, this has a concrete implication: monitoring velocity RMSE alone is insufficient, and a staged training curriculum that decouples the forward and inverse problems during early epochs may be necessary.

More broadly, our results reinforce that PINN training enhancements cannot be treated as interchangeable plug-ins. Each interacts with the problem structure in ways that theoretical motivation alone does not predict. Systematic ablation on the specific problem at hand remains essential.

A natural next step is to combine the two successful improvements. Because sinusoidal activations and adaptive collocation operate on orthogonal aspects of the training (network representational capacity versus loss sampling efficiency), we hypothesize that their benefits should compose nearly additively. The NTK perspective also suggests a principled Fourier feature bandwidth: $\sigma \approx fL/\alpha_0 \approx 9.3$ for our configuration, which would test whether the encoding's failure is fundamental or merely a tuning issue.

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